

COVER STORY

India's Silent Lipid Epidemic: A Ticking Cardiovascular Clock



ICMR–INDIAB data highlights a major burden: 79% of Indian adults have ≥1 lipid abnormality—most remain undetected and untreated.¹

India is at a critical point in its lipid landscape, with emerging evidence showing a distinct and highly prevalent pattern of dyslipidemia among Asian Indians. The ICMR–INDIAB Phase I study (adults aged 20–90 years, across regions) reported a substantial burden of lipid abnormalities:¹

13.9% Hypercholesterolemia ¹	72.3% Low HDL-C ¹
29.5% Hypertriglyceridemia ¹	11.8% High LDL-C ¹

"In Indian adults, the most common dyslipidemia pattern is low HDL-C followed by high triglycerides — atherogenic dyslipidemia is India's defining challenge."

CARDIOVASCULAR HEALTH

The Liver–Heart Highway: NAFLD Raises CVD Risk by 64%²

A landmark meta-analysis proves NAFLD is not just a liver disease — it is a potent cardiovascular time bomb



When the liver accumulates fat, the heart silently suffers. The data are now irrefutable.

Targher et al. conducted the most comprehensive meta-analysis to date — incorporating 16 studies, 34,043 adults, and ~2,600 CVD events over a median follow-up of 6.9 years.²

NAFLD & CVD Risk: Odds Ratios at a Glance ²	
Any NAFLD vs. No NAFLD	OR 1.64
Non-fatal CVD Events	OR 2.52
Severe NAFLD — CVD Mortality	OR 3.28
Severe NAFLD — Fatal+Non-fatal	OR 2.58

OR = Odds Ratio; Random effects model; 95% CI statistically significant

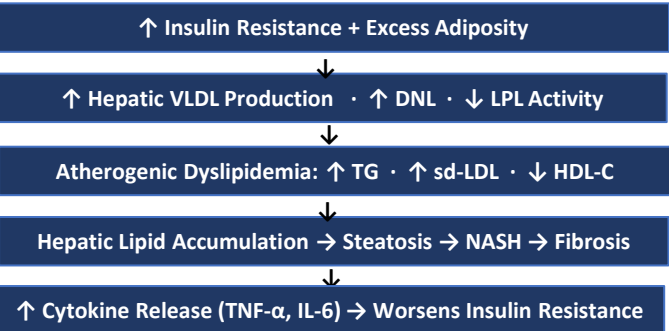
MASLD / LIVER HEALTH

The Atherogenic Dyslipidemia–MASLD Vicious Cycle

60–70% of MASLD patients carry dyslipidemia. Lipids worsen the liver; the diseased liver worsens the lipids.³



The Atherogenic Dyslipidemia–MASLD Vicious Cycle⁴



CYCLE REPEATS — Escalating ASCVD Risk

MASLD Key Facts

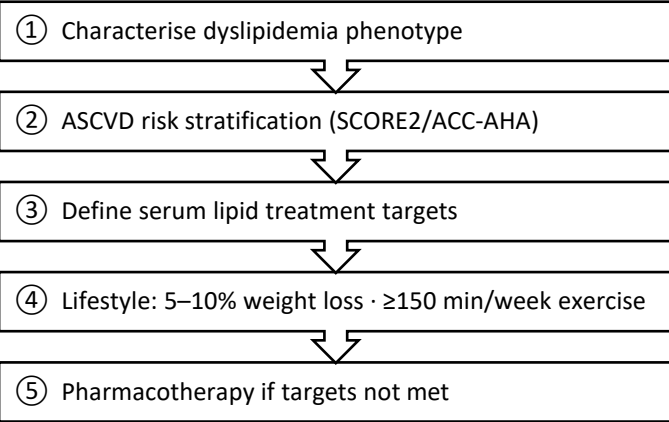
- Affects up to **25%** of the global adult population⁴
- Dyslipidemia in **60–70%** of MASLD patients³
- Reduces life expectancy by **2.6 years** on average³

"Atherogenic dyslipidemia and NAFLD share a bidirectional relationship and are both major driving forces of ASCVD — early detection and adequate treatment are urgently warranted."

MANAGEMENT INSIGHTS

Treating Dyslipidemia in NAFLD:

The 5-Step Roadmap⁴



DUAL PPAR SCIENCE

The Dual PPAR Advantage: Targeting Both Lipids and Glucose — Where Other Agents Fall Short

Saroglitazar's unique dual PPAR-α/γ mechanism addresses the very metabolic pathways that drive atherogenic dyslipidemia in T2DM¹

Why Dual PPAR-α/γ Matters¹

- **PPAR-α activation:** ↑ LPL activity, ↓ ApoC-III → ↓ TG, ↑ HDL-C
- **PPAR-γ activation:** ↑ Adiponectin, glucose transporters → ↑ insulin sensitivity
- **Dual PPAR-α/γ:** Simultaneously addresses hypertriglyceridaemia and insulin resistance.

CLINICAL TRIALS

PRESS V & VI:



–45% TG reduction, Saroglitazar 4 mg vs. Pioglitazone (PRESS V) ¹	–46.7% TG reduction vs. Placebo (PRESS VI) ¹
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PRESS V Additional Benefits with Saroglitazar 4 mg¹

- ↓ VLDL-C by **45.5%** ↓ Total cholesterol by **7.7%**
- ↓ ApoB by **10.9%** ↓ LDL-C by **5%**
- **No edema, no weight gain** — unlike pioglitazone
- No significant adverse cardiac or hepatic events

NATIONAL LIPID DAY 2026: AWARENESS

- ❑ Screen all T2DM and MetS patients for dyslipidemia and fatty liver¹
- ❑ Assess non-HDL-C and ApoB — not LDL-C alone^{1,2}
- ❑ Recognise low HDL-C + high TG as India's dominant atherogenic phenotype¹
- ❑ Educate patients: Dyslipidemia is often silent until a cardiac or hepatic event
- ❑ Pursue integrative CV risk + liver imaging in high-risk patients^{2,4}

National Lipid Day 2026 — Screen Every At-Risk Patient. Treat Beyond LDL. Protect the Liver. Save the Heart.

ACC-AHA – American College of Cardiology / American Heart Association, ApoB – Apolipoprotein B, ASCVD – Atherosclerotic Cardiovascular Disease, CVD – Cardiovascular Disease, DNL – De Novo Lipogenesis, HDL-C – High-Density Lipoprotein Cholesterol, ICMR–INDIAB – Indian Council of Medical Research – India Diabetes Study, LDL-C – Low-Density Lipoprotein Cholesterol, LPL – Lipoprotein Lipase, MASLD – Metabolic Dysfunction–Associated Steatotic Liver Disease, NAFLD – Non-Alcoholic Fatty Liver Disease, NASH – Non-Alcoholic Steatohepatitis, PPAR-α/γ – Peroxisome Proliferator-Activated Receptor Alpha/Gamma, sd-LDL – Small Dense Low-Density Lipoprotein, T2DM – Type 2 Diabetes Mellitus, TG – Triglycerides, VLDL – Very Low-Density Lipoprotein

References:
1. Joshi SR. Saroglitazar for the treatment of dyslipidemia in diabetic patients. Expert Opin Pharmacother. 2015 Mar;16(4):597-606. doi: 10.1517/14656566.2015.1009894. PMID: 25674933.
2. Targher G, Byrne CD, Lonardo A, Zoppini G, Barbui C. Non-alcoholic fatty liver disease and risk of incident cardiovascular disease: A meta-analysis. J Hepatol. 2016 Sep;65(3):589-600. doi: 10.1016/j.jhep.2016.05.013. Epub 2016 May 17. PMID: 27212244.
3. National Lipid Association. Practical pearls: dyslipidemia management in metabolic dysfunction–associated steatotic liver disease (MASLD). LipidSpin. 2024 Spring.
4. Martin A, Lang S, Goesser T, Demir M, Steffen HM, Kasper P. Management of Dyslipidemia in Patients with Non-Alcoholic Fatty Liver Disease. Curr Atheroscler Rep. 2022 Jul;24(7):533-546. doi: 10.1007/s11883-022-01028-4. Epub 2022 May 4. PMID: 35507279; PMCID: PMC9236990.

